MD. NAZMUL HASAN SHEUM

Senior Backend Engineer | Software Architecture

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 7+ years of experience building and maintaining production web applications using PHP, Laravel, CodeIgniter, and MySQL. Skilled in backend architecture, REST API design, database performance optimization, application security, and cloud deployments. Experienced in translating business requirements into practical technical solutions, collaborating directly with customers, and driving implementation from design through release. Proven ability to lead small engineering teams, conduct code reviews, mentor developers, and deliver reliable, maintainable software.

TECHNICAL SKILLS

Backend: PHP 8.x, Laravel, CodeIgniter, Python (Flask), JavaScript (ES6+), RESTful APIs, API Integration

Architecture & Design: System Design, Clean Architecture, Modular Monolith, SOLID Principles

Database Engineering & Performance: MySQL, Redis, Database Design, Indexing Strategies, Query Optimization, Execution Plan Analysis (EXPLAIN), Caching Strategies, Performance Profiling

Security & Reliability: OWASP Top 10, Authentication & Authorization, Input Validation, Rate Limiting, Idempotency, Retry Strategies, Asynchronous Processing

Cloud & DevOps: AWS (EC2, S3, RDS, Auto Scaling, CodePipeline), Google Cloud Platform (GCP), Docker, Linux, CI/CD

Testing & Code Quality: PHPUnit, Unit & Feature Testing, SonarQube, Code Reviews, JMeter

Frontend & Templating: Blade, Jinja, Vue.js, AngularJS, Bootstrap 5, Tailwind CSS

PROFESSIONAL EXPERIENCE

Senior Software Engineer **Aug 2023 – Present**

Brain Station 23 PLC

- Develop and maintain enterprise applications for a major telecommunications client.
- Collaborate directly with client stakeholders to gather requirements, propose technical solutions, and guide implementation from design through delivery.
- Design and maintain backend modules and REST APIs, applying appropriate architectural patterns and SOLID principles.
- Optimize MySQL queries and application performance through execution-plan analysis, indexing, and caching where appropriate.
- Introduced unit and feature testing practices, improving release reliability.
- Conduct code reviews, mentor developers, and help maintain code quality, security, and release reliability.
- Improve automated testing and address code-quality and security findings using PHPUnit and SonarQube.

Software Engineer (Technical Lead responsibilities from 2021) **Jul 2020 – Jul 2023**

Oneixchange LTD, Singapore (Remote)

- Led backend development and architected RESTful services for business-critical web and mobile applications, delivering multiple projects for Singapore government clients, including MOE Singapore and Dunearn Secondary School.
- Supervised a 5-member engineering team, providing code review, architecture guidance, and performance optimization mentorship.
- Implemented CI/CD workflows and collaborated with DevOps on AWS and GCP infrastructure.

Junior Programmer **Sep 2017 – Apr 2019**

Freelance IT Lab

- Developed backend features in PHP (CodeIgniter) for multiple client projects, including e-commerce platforms.
- Contributed to performance improvements, bug fixes, and delivery of production-ready applications.

KEY PROJECTS

Fixed Asset Management | *Enterprise Internal System, Telecom* | *Team of 14* | *Senior Backend Engineer*

- Designed asset lifecycle modules with complex relational data modeling, built as a modular monolith serving multiple user domains on a large database.
- Optimized the database schema, indexing strategy, and performance bottlenecks to improve query efficiency.
- Integrated third-party enterprise APIs with resilient, fault-tolerant handling.
- Enforced SonarQube compliance and completed security remediation.

Relief Teacher System | *Dunearn Secondary School, Singapore* | *Team of 8* | *Lead Backend Engineer*

- Architected an API-first backend supporting AngularJS web and mobile platforms.
- Implemented rule-based allocation logic with a modular HMVC architecture for maintainability.
- Optimized database queries and caching for consistent performance at a regular peak load of 500 concurrent users.

STEM Calendar | *MOE Singapore* | *Team of 5* | *Lead Backend Engineer*

- Designed backend services supporting public event listings and registrations.
- Managed AWS-based deployment and database optimization.

EDUCATION

B.Sc. in Computer Science and Engineering **2012 – 2017**
Jashore University of Science and Technology (JUST)

ACHIEVEMENTS & CERTIFICATIONS

- Star Performer (Performance Excellence Award), Brain Station 23 PLC, SBU PNC & TELCO, 2024 and 2026.
- Delivered enterprise-grade systems used by 1,000+ internal users.
- First Place, Intra-university Programming Contest (2013).
- HackerRank Python Gold Badge and SQL (Intermediate) Certification.